MIL-F-48033 (MU)
22 February 1972
(see 6.4)
SUPERSEDING
PA-PD-2628A
27 March 1967

MILITARY SPECIFICATION
FUZE, POINT INITIATING, BASE DETONATING M539, LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of fuze designated as Fuze, Point Initiating, Base Detonating, M539.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification as specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

9204104 - Box, Packing, Ammunition for Fuze, PIBD, M539
9204105 - Box, Fiber, Packing Ammunition for Fuze, PIBD, M539
9212208 - Fuze, PIBD, M539, Loading Assembly

PUBLICATIONS

ARMY

ET-9212208 - Equipment Tabulation

FSC: 1390

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Fuze assembly.-The fuze assembly shall comply with all requirements specified on drawing (dwg.) 9212208, and with all requirements specified in applicable specification.

3.3 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined. All packing components shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an identification and date of changes.

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e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Each lot shall contain:

a. Fuze parts from one lot interfix number from one manufacturer.

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b. RDX for lead charges from no more than two consecutive lots with the same lot interfix number from one manufacturer.

c. RDX for booster pellets from no more than two consecutive lots with the same lot interfix number from one manufacturer.

4.2.2 Examination.-Inspection for Critical defects (and Major defects, when so specified,) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Booster Pellet (see dwg. 9214669 covering a detail of dwg. 9212208).

Categories	Defects	Method of Inspection	Code No.
Critical:	Non defined		
Major:	AQL 0.40 percent		
101.	Diameter	Gage	01001
102.	Thickness	Gage	01002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.3)	Visual	01003

4.2.2.2 Booster assembly prior to assembling disc (see dwg. 9214661 covering a detail of dwg. 9212208)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.3)	Visual	02001

4.2.2.3 Booster assembly (see dwg. 9214661 covering a detail of dwg. 9212208)

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Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.5 percent		
201.	Lacquer seal missing	Visual	03001
202.	Crip not full 360 degrees	Visual	03002
203.	Evidence of poor workmanship (see 3.3)	Visual	03003

4.2.2.4 Lead cup assembly (see dwg. 9214668 covering a detail of dwg. 9212208)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Length, max	Gage	04001
102.	Face of charge excessively deep ...	Gage	04002
Minor:	AQL 1.5 percent		
201.	Face of charge above flush	Gage	04003
202.	Varnish missing from face of charge	Visual	04004
203.	Charge cracked	Visual	04005

4.2.2.5 Fuze prior to assembling booster (see dwg. 9212208)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Lead missing	Visual	05001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.3)	Visual	05002

4.2.2.6 Booster assembly, prior to assembling to fuze (see dwg. 9212208)

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Categories	Defects	Method of Inspection	Code NO.
Critical: None defined			
Major: None defined			
Minor:	AQL 0.65 percent		
201.	Thread sealant missing.....	Visual	06001

4.2.2.7 Fuze assembly (see dwg. 9212208)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.5 percent		
201.	Booster assembly fails to withstand application of minimum assembly torque	Gage	07001
202.	Length, max.....	Gage	07002
203.	Marking misleading or unidentifiable	Visual	07003
204.	Evidence of poor workmanship (see 3.3)	Visual	07004

4.2.2.8 Box, fiber, packing, prior to sealing (see dwg. 9204105)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.5 percent		
201.	Any packing material missing	Visual	08001
202.	Separator damaged to the extent fuzes touch each other	Visual	08002
203.	Component missing	Visual	08003

4.2.2.9 Box, fiber packing, after sealing (see dwg. 9204105)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			

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Major: None defined

Minor:		AQL 2.5 percent	
201.	Carton damaged to the extent that contents are exposed	Visual	09001
202.	Sealing strip missing or insecure	Visual	09002
203.	Contents loose	Visual	09003
204.	Marking, misleading or unidentifiable	Visual	09004
205.	Evidence of poor workmanship (see 3.3)	Visual	09005

4.2.2.10 Sealed barrier bag (see dwg. 9204105)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Seal improper or incomplete	Visual	10001
102.	Bag, punctured, torn or cut.....	Visual	10002
Minor:	AQL 1.0 percent		
201.	Marking misleading or unidentifiable	Visual	10003
202.	Evidence of poor workmanship (see 3.3)	Visual	10004

4.2.2.11 Packing box, prior to sealing (see dwg. 9204104)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 0.65 percent		
201.	Carton missing	Visual	11001

4.2.2.12 Packing box, prior to sealing (see dwg. 9204104)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		

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101.	Box broken, board loose or protruding nail	Visual	12001
102.	Strapping missing broken or loose	Visual	12002
103.	Marking misleading or unidentifiable ..	Visual	12003
Minor:	AQL 1.00 percent		
201.	Strapping improperly engaged	Visual	12004
202.	Contents loose	Manual	12005

4.2.3 Testing

4.2.3.1 Weight of booster pellet (see dwg. 9214669) covering a detail of dwg. 9212208 (Major defect, Code NO. 01004).-Thirty sample booster pellets shall be selected from each lot for this test. If one or more samples does not meet the applicable requirement, the lot shall be rejected. Equipment used for this test shall be in accordance with 4.2.4.

4.2.3.2 Continuity of disarmed shorting circuit (see dwg. 9212208) (Critical defect, Code No. 07005).-This test shall be conducted 100 percent. Any assembly that fails to comply with the applicable requirement shall be removed from the lot. Testing shall be in accordance with 4.3.1.

4.2.3.3 X-ray of Loaded fuze Ballistic Samples.-All fully loaded fuzes shall be x-rayed prior to assembly into complete rounds for proving ground acceptance test of fuze metal parts to insure that booster and lead charges are present and that the rotor is in the unarmed position. Rotor not in unarmed position - Critical Defect - Code No. 07006, Lead or booster charge missing - Major Defect - Code No. 07007. Equipment used for this test shall be in accordance with 4.2.4.

4.2.4 Inspection Equipment.-Equipment Tabulation Number ET-9212208 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection equipment.-The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel

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for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test Methods and Procedures

4.3.1 Continuity of disarmed shorting circuit.-This test shall be conducted on assembled fuzes using an ohmmeter having a short circuit current of not more that 20 microamperes. Appropriate safety precautions shall be taken during this test using equipment in accordance with 4.2.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C.-The loaded fuzes shall be preserved and packaged in accordance with dwg. 9204105.

5.2 Packing

5.2.1 Level C.-The fiber boxes shall be packed in accordance with dwg. 9204104.

5.3 Marking.-Marking shall be in accordance with dwg. 9204104 and dwg. 9204105.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes

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as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration intructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.5 Submission of contractor designs. - All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-F. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 This specification includes the requirements of PA-PD-2628A dated 27 March 1967.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number 1390-A-003

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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Dover, New Jersey 07801

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